

Ch 13: Our Environment

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SECTION A - ECOSYSTEM, ENVIRONMENT AND ITS COMPONENTS

1 WHAT IS ENVIRONMENT?

The environment includes all the surroundings—living and non-living things—that influence the life and activities of organisms.

2 WHAT IS AN ECOSYSTEM?

An ecosystem is a functional unit of nature where living organisms (biotic) interact with non-living surroundings (abiotic) and exchange matter and energy among themselves.

- It is self-sustaining and maintains a balance.

3 COMPONENTS OF AN ECOSYSTEM

An ecosystem has two main components:

- Biotic components** – living organisms.
- Abiotic components** – non-living physical and chemical factors.

4 BIOTIC COMPONENTS

Living components of an ecosystem are classified into three groups:

① Producers (Autotrophs)

Organisms that make their own food using sunlight, water, CO_2 and minerals.

Examples: Green plants, algae, cyanobacteria.

② Consumers (Heterotrophs)

Organisms that cannot make their own food and depend on other organisms for food.

Examples: Herbivores – Cow, Deer, Rabbit
Carnivores – Lion, Tiger, Snake
Omnivores – Human, Bear, Crow

③ Decomposers

Organisms that break down dead plants, animals and wastes into simpler substances and return nutrients to the soil.

Examples: Bacteria, Fungi, Earthworms.

5 ABIOTIC FACTORS

Non-living components that affect the living organisms in an ecosystem.

- Sunlight
- Air (O_2 , CO_2)
- Water
- Soil
- Temperature
- Minerals/Nutrients

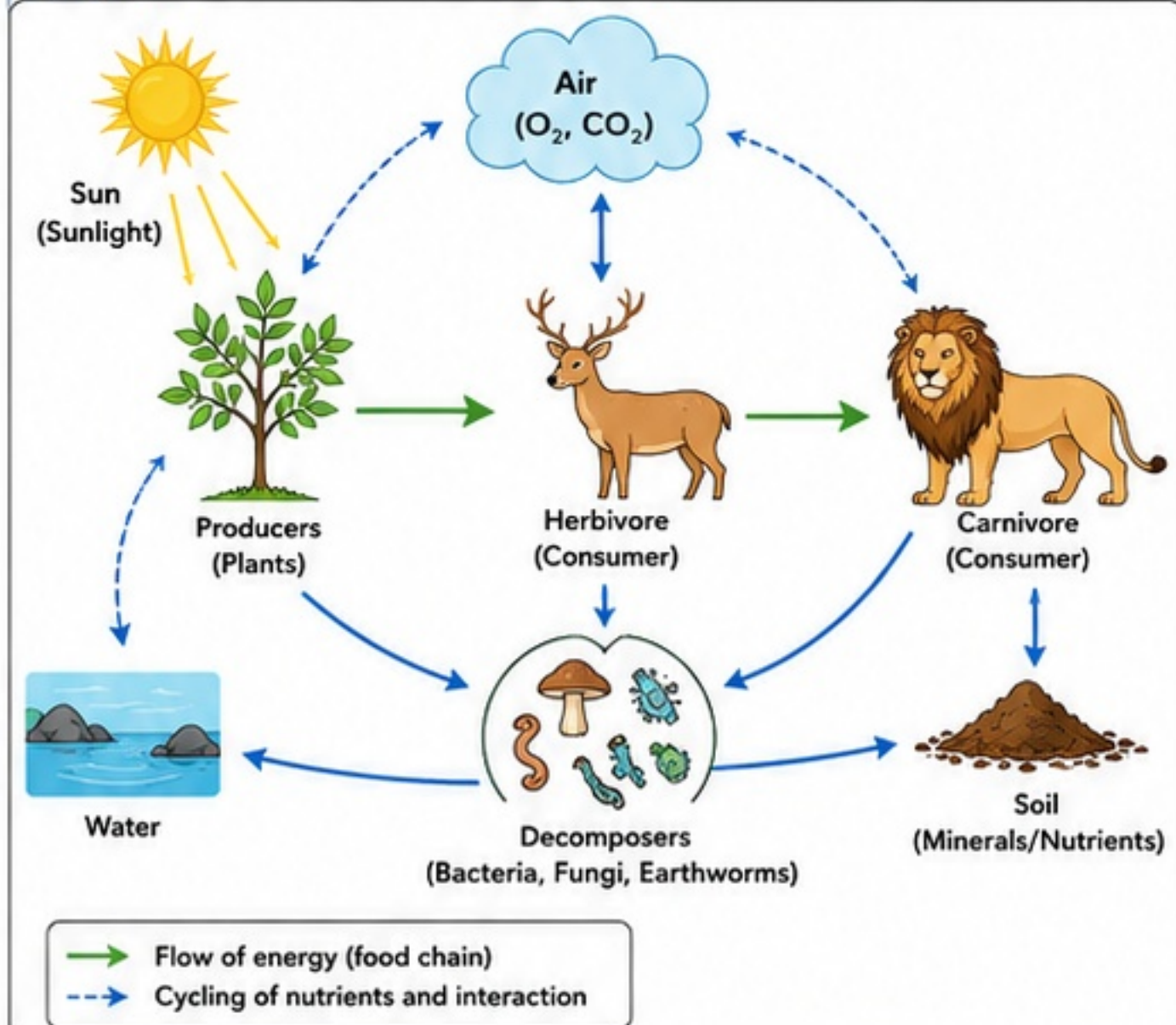
IMPORTANT TERMS

Biotic	: Living components of an ecosystem.
Abiotic	: Non-living components of an ecosystem.
Producer	: Organism that makes its own food.
Consumer	: Organism that depends on others for food.
Decomposer	: Organism that breaks down dead matter.

6 INTERDEPENDENCE IN AN ECOSYSTEM

- All components of an ecosystem are interdependent. A change in one component affects others.
- Flow of energy and cycling of nutrients link living and non-living components.
- Balance and harmony are essential for the stability of an ecosystem.

7 ECOSYSTEM – CONCEPT MAP



8 ENVIRONMENT VS ECOSYSTEM

Basis	Environment	Ecosystem
Meaning	Includes all surroundings—biotic and abiotic—of a living organism.	A functional unit where biotic and abiotic components interact.
Scope	Very broad; includes many ecosystems.	Limited to a specific area (e.g., pond, forest).
Nature	May or may not involve interaction.	Always involves interaction among components.
Function	Does not necessarily function as a unit.	Functions as a self-sustaining unit of nature.
Example	Earth, atmosphere, biosphere.	Pond ecosystem, forest ecosystem, grassland ecosystem.

QUICK RECALL

- Environment = Surroundings (living + non-living)
- Ecosystem = Functional unit of nature
- Biotic = Producers, Consumers, Decomposers
- Abiotic = Sunlight, Air, Water, Soil, Temperature, Minerals

KEY IDEA

Ecosystem works as a team: biotic and abiotic components interact continuously, exchange matter and energy, and maintain balance.

EXAM TIPS

- Define environment and ecosystem clearly.
- Classify biotic components correctly.
- Give examples for each component.
- Draw neat labelled diagrams for ecosystem.
- Use comparison tables in answers.

CENTUM FOCUS

- Learn definitions word-for-word.
- Understand interdependence deeply.
- Practice diagrams & flow charts.
- Write points in systematic order.
- Revise tables and terms daily.



Understand nature, respect it, protect it — because a balanced environment ensures a better future!



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SECTION B – FOOD CHAIN, TROPHIC LEVELS AND FOOD WEB

1 MEANING OF FOOD CHAIN

A food chain is a linear sequence of organisms in which each organism is eaten by the next one in the sequence for the transfer of energy and nutrients.

2 TYPES OF FOOD CHAINS

① Grazing Food Chain

- Begins with living green plants (producers) and passes energy through herbivores to carnivores.

② Detritus Food Chain

- Begins with dead organic matter (detritus) and passes energy through decomposers and detritivores to higher consumers.

3 TROPHIC LEVEL – MEANING

A trophic level is each step or position in a food chain where energy is transferred from one organism to another.

4 MAIN TROPHIC LEVELS

① Producers (Autotrophs)

Make their own food using sunlight, water, CO₂ and minerals (e.g., green plants, phytoplankton).

② Primary Consumers (Herbivores)

Feed on producers (e.g., deer, grasshopper, rabbit, zooplankton).

③ Secondary Consumers (Carnivores)

Feed on primary consumers (e.g., frog, snake, small fish).

④ Tertiary Consumers (Top Carnivores)

Feed on secondary consumers (e.g., eagle, lion, large fish, human).

5 EXAMPLE FOOD CHAINS

① Terrestrial (Simple):

Grass → Deer → Lion

② Terrestrial (Complex):

Grass → Grasshopper → Frog → Snake → Eagle

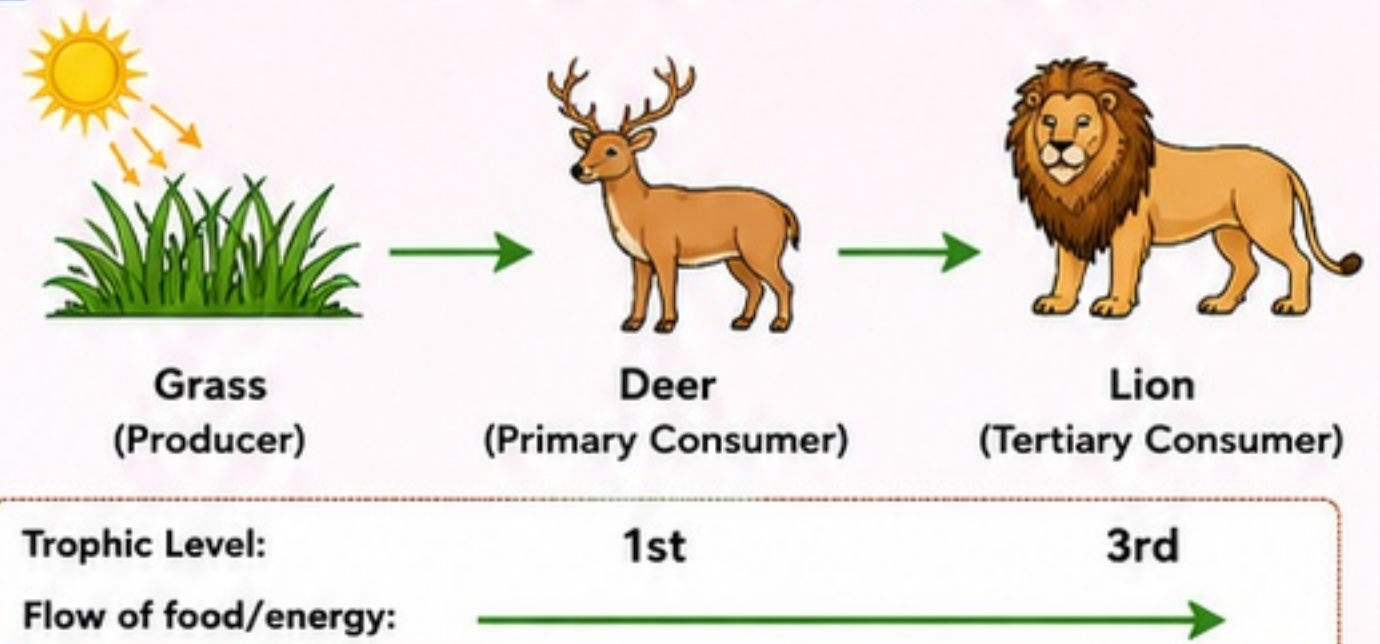
③ Aquatic:

Phytoplankton → Zooplankton → Fish → Large Fish → Bird / Human

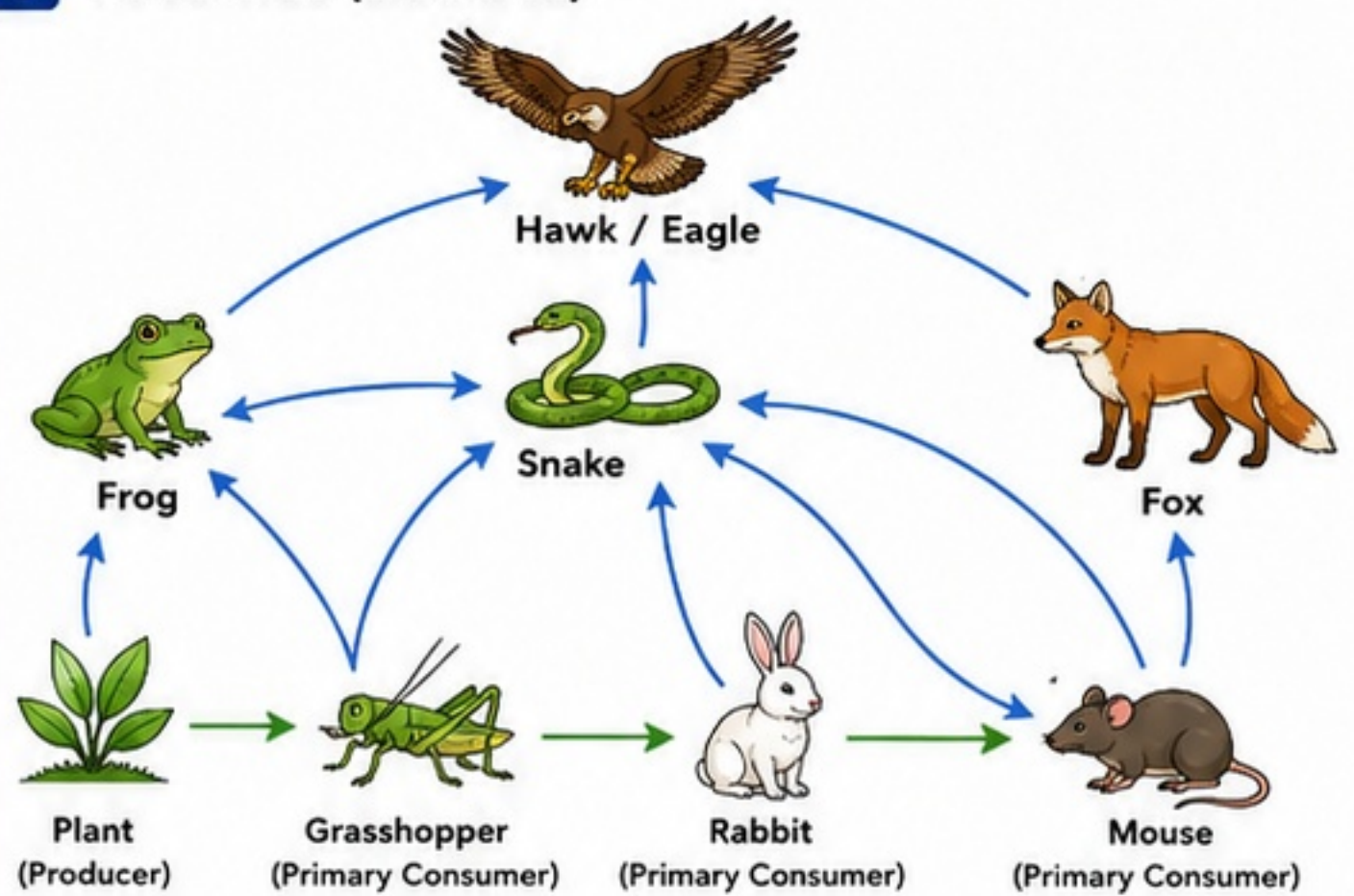
6 WHY FOOD CHAINS ARE GENERALLY SHORT

- At each transfer of energy, a large amount of energy is lost as heat.
- Only about 10% of energy is passed to the next level (10% law).
- Insufficient energy remains after 4–5 steps to support more levels.
- Hence, most food chains have only 3–5 links.

7 FOOD CHAIN WITH TROPHIC LEVELS (EXAMPLE)



8 FOOD WEB (EXAMPLE)



9 FOOD CHAIN vs FOOD WEB

Basis	Food Chain	Food Web
Structure	Linear sequence	Network of linked food chains
Pathway	Single pathway of energy flow	Multiple pathways of energy flow
Complexity	Simple	Complex
Stability	Less stable; more affected by changes	More stable; alternative food sources available
Example	Grass → Deer → Lion	Plant → Grasshopper → Frog → Snake → Hawk; Plant → Mouse → Snake → Hawk (many links)

10 INTERDEPENDENCE AND STABILITY THROUGH FOOD WEBS

- Food webs show that organisms are interdependent.
- If one organism decreases or disappears, others can still survive using alternative food sources.
- This interdependence maintains balance and stability in ecosystems.

QUICK RECALL

- Food chain = Linear path of energy.
- 10% law = Only 10% energy passes to next level.
- Food chains are short (3–5 links).
- Food web = Network of many food chains.
- Food webs increase stability.

KEY IDEA

Energy flows from producers to top consumers through trophic levels. Food webs provide multiple pathways of energy flow, ensuring interdependence and stability in nature.

EXAM TIPS

- Draw food chains with arrows showing energy flow.
- Identify trophic levels correctly.
- Give at least one grazing and one aquatic food chain example.
- For food web questions, show interconnections clearly.

CENTUM FOCUS

- Learn definitions word-for-word.
- Differentiate food chain and food web clearly.
- Practise diagrams and examples.
- Write precise, point-wise answers.
- Revise tables and key terms daily.



Understand how energy flows, respect life, protect balance — because every link in nature matters!



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SECTION C – ENERGY FLOW IN AN ECOSYSTEM AND THE 10% LAW

1 ENERGY ENTERS ECOSYSTEM FROM THE SUN

The ultimate source of energy for almost all ecosystems is the Sun. Solar energy drives all life processes on Earth.



2 PRODUCERS TRAP SOLAR ENERGY

Green plants (producers) trap sunlight using chlorophyll and convert it into chemical energy by photosynthesis. This stored energy is the starting point of every food chain.



3 ENERGY FLOW IS UNIDIRECTIONAL

- Energy flows from one trophic level to the next.
- It does not cycle back to the previous level.
- Lost energy is not reused by the same organism.

4 WHY ENERGY DECREASES AT EACH LEVEL

- A large part of energy is used in respiration, movement, growth and reproduction.
- Heat energy is lost to the surroundings.
- Some energy is lost in excretory products and uneaten parts.

5 LINDEMAN'S 10% LAW

Raymond Lindeman proposed that only about 10% of the energy at one trophic level is transferred to the next level. About 90% is lost as heat or used in life processes.

6 SAMPLE NUMERICAL / EXAMPLE

If producers (plants) have 10,000 J of energy:

- Primary consumers get 10% → 1,000 J
 - Secondary consumers get 10% of 1,000 J → 100 J
 - Tertiary consumers get 10% of 100 J → 10 J
- Energy decreases ten times at each step.

7 WHY LESS ENERGY AT HIGHER LEVELS?

- Most energy used up in metabolic activities.
- Energy lost as heat to the environment.
- Energy lost in waste and uneaten materials.
- Only a small fraction stored in new biomass.

8 WHY TOP CARNIVORES ARE FEWER?

- Very little energy is available at higher levels.
- Insufficient energy cannot support large populations.
- Hence, top carnivores are few in number.

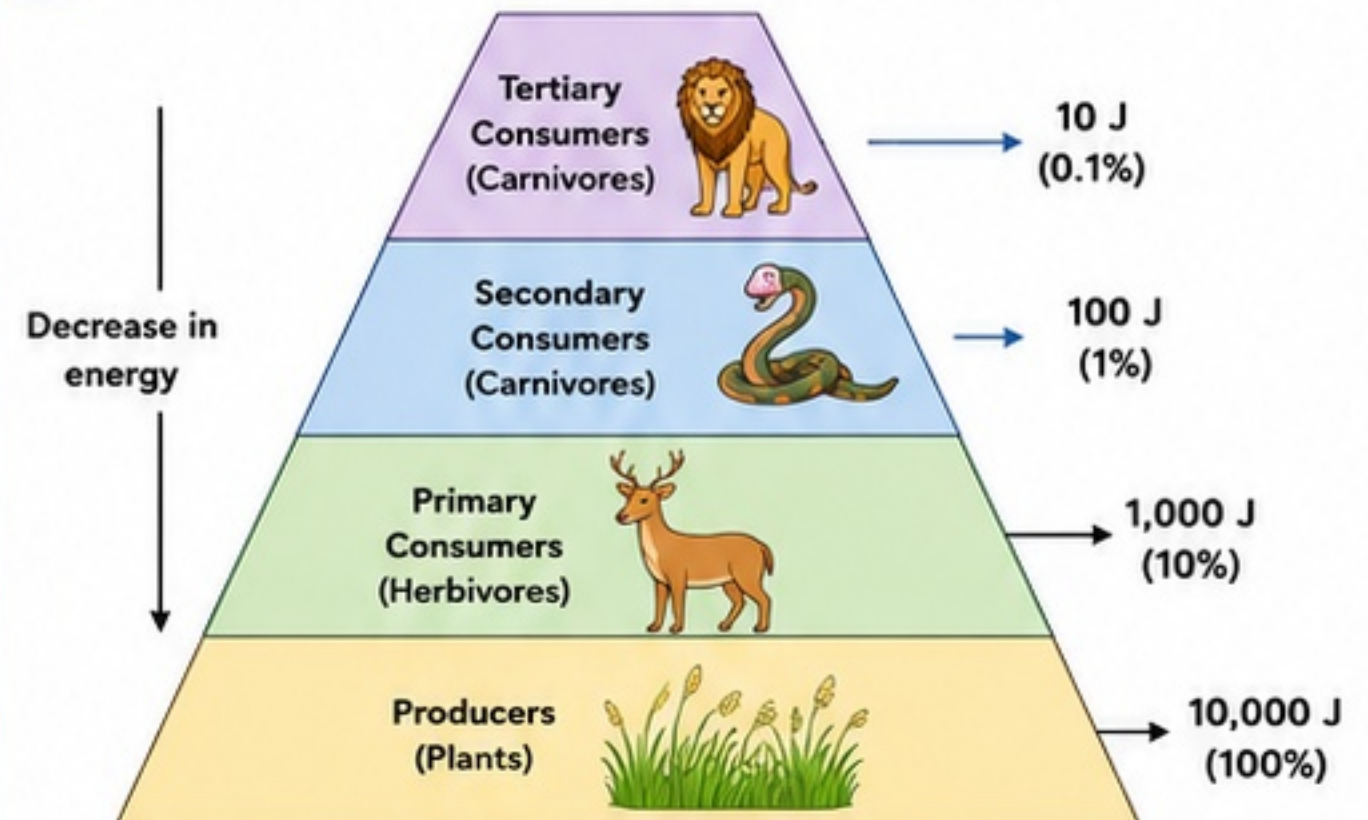
9 ENERGY PYRAMID / ECOLOGICAL PYRAMID

A diagrammatic representation showing decrease in energy at successive trophic levels. It is always upright due to continuous loss of energy at each level.

10 IMPORTANCE OF UNDERSTANDING ENERGY FLOW

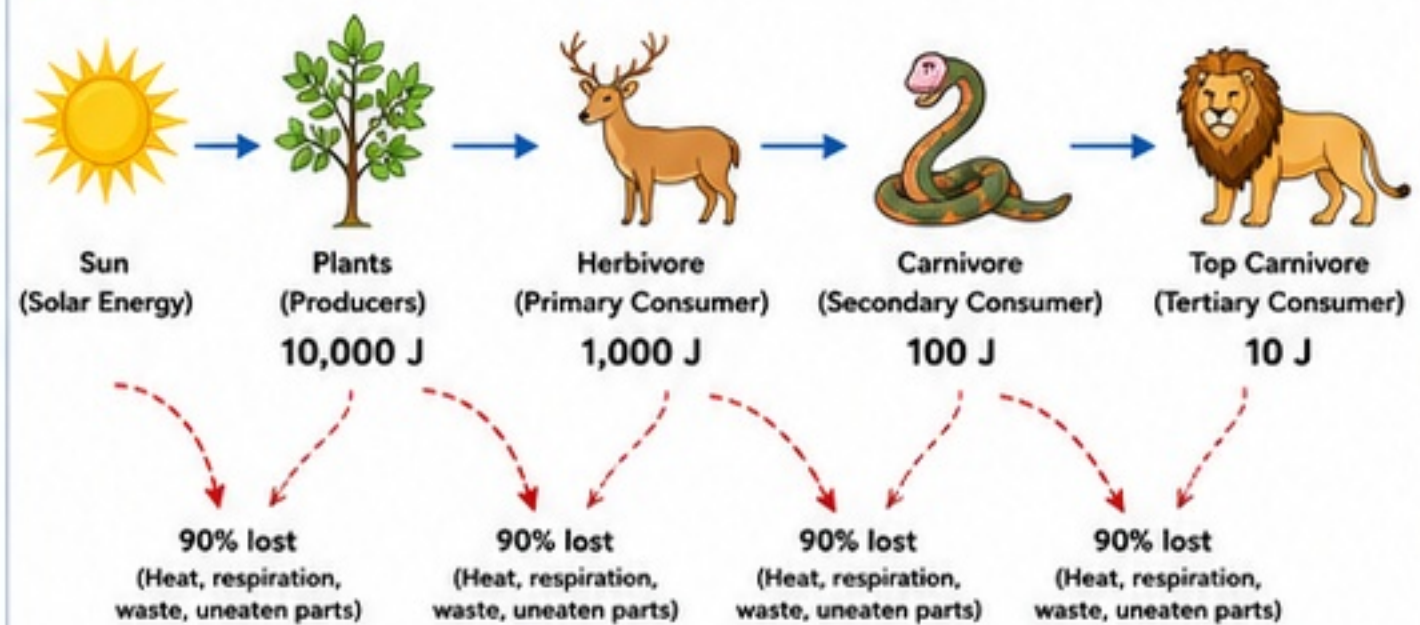
- Helps us understand ecosystem functioning.
- Explains food chain length and population control.
- Guides conservation and sustainable use of resources.
- Helps maintain ecological balance and biodiversity.

9 ENERGY PYRAMID (ECOLOGICAL PYRAMID)



Only about 10% of energy is transferred to the next trophic level.

6 SAMPLE ENERGY FLOW (STEP FLOW CHART)



Energy flows in one direction and decreases at every step.

TROPHIC LEVELS AND ENERGY AVAILABILITY

Trophic Level	Organism Type	Example	Approximate Energy Available
1. Producers	Autotrophs (Plants, Algae)	Green plants, phytoplankton	10,000 J (100%)
2. Primary Consumers	Herbivores	Deer, Rabbit, Grasshopper	1,000 J (10%)
3. Secondary Consumers	Carnivores	Snake, Frog, Small Birds	100 J (1%)
4. Tertiary Consumers	Top Carnivores	Eagle, Lion	10 J (0.1%)



WHY FOOD CHAINS HAVE ONLY 3–5 TROPHIC LEVELS?

- Due to the 10% law, energy reduces tenfold at each level.
- After 4–5 levels, too little energy remains to support life.
- Hence, food chains are short (usually 3 to 5 trophic levels).



QUICK RECALL

- Sun is the ultimate energy source.
- Producers trap energy via photosynthesis.
- Energy flows in one direction.
- Only ~10% energy passed on.
- Energy pyramids are always upright.



KEY IDEA

Energy enters as sunlight, is trapped by producers, and flows from one trophic level to the next with ~10% transfer efficiency, decreasing at each step.



EXAM TIPS

- Learn the 10% law and its explanation clearly.
- Draw neat energy pyramids with labels and values.
- Learn numerical examples for better marks.
- Use comparison tables.



CENTUM FOCUS

- Learn definitions word-for-word.
- Practice diagrams and step-flow charts regularly.
- Write points in systematic order.
- Revise tables and key numbers daily.

★ Understand energy flow, respect nature, conserve energy — because less waste means a better future! ★

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SECTION D – DECOMPOSERS, NUTRIENT CYCLING AND ECOLOGICAL BALANCE

1 MEANING OF DECOMPOSERS

Decomposers are organisms that break down dead plants, animals and organic wastes into simpler substances. They feed on dead matter and return nutrients to the environment.

2 EXAMPLES OF DECOMPOSERS

- **Bacteria** – microscopic organisms that decompose dead matter.
- **Fungi** – moulds and mushrooms that break down dead plants and animals.
- **Earthworms** – pull dead leaves into soil and mix them with soil, helping decomposition.

3 ROLE OF DECOMPOSERS

- Break down complex organic matter into simple substances.
- Convert dead plants, animals and wastes into humus, minerals and gases.
- Help in cleaning the environment.

4 RECYCLING OF NUTRIENTS

The nutrients released during decomposition are absorbed by soil or water. These nutrients are again taken up by plants and used to make food. Thus, nutrients are recycled in nature.

5 IMPORTANCE OF DECOMPOSITION

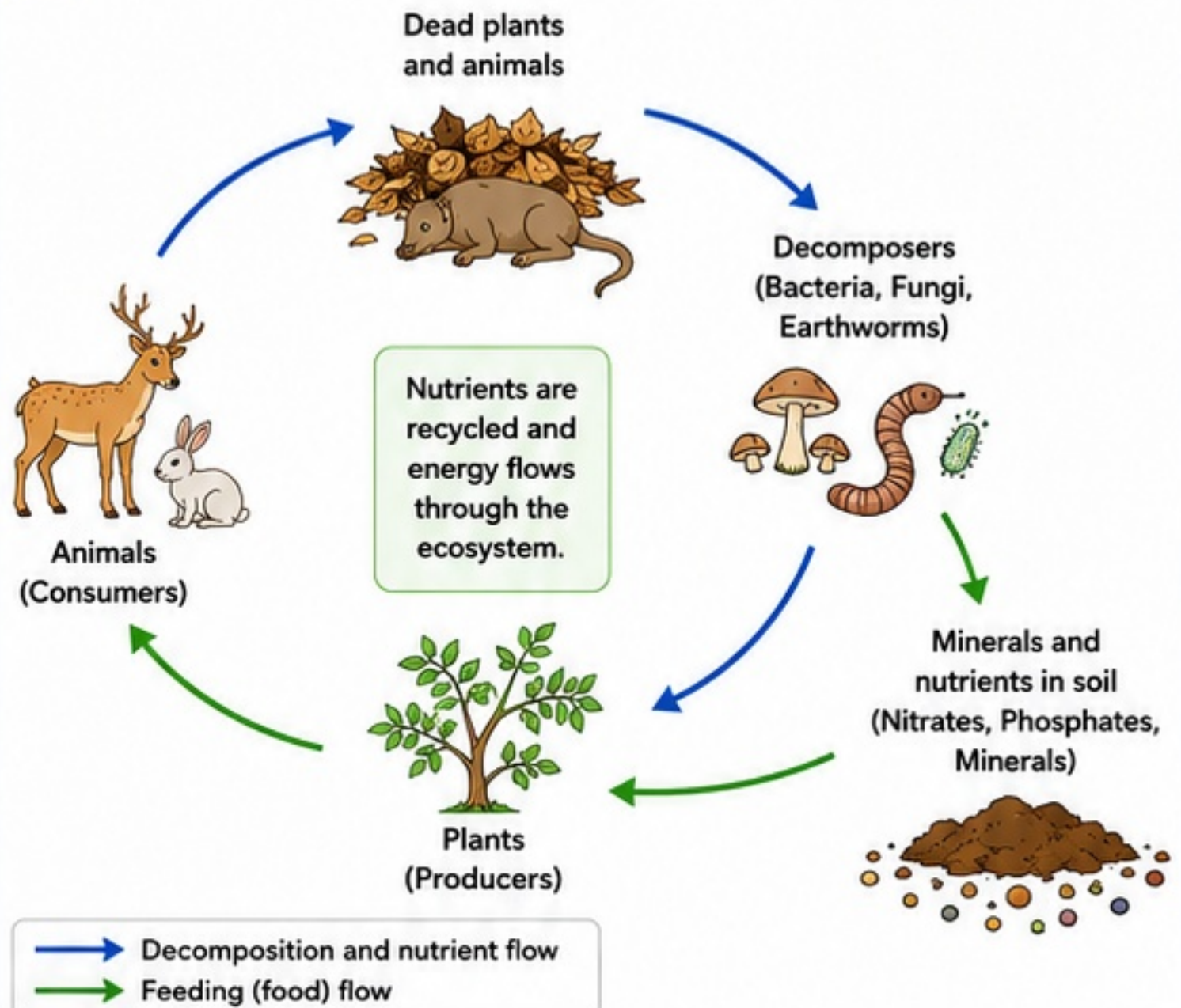
- Keeps the environment clean by removing dead organisms and wastes.
- Prevents accumulation of organic matter.
- Maintains continuity of life by returning nutrients for new life.

6 NUTRIENT CYCLES OVERVIEW

Nutrients and elements go through continuous cycles in nature. The main cycles are:

 Carbon Cycle	CO ₂ in air → Plants use in photosynthesis → Animals eat plants → CO ₂ returned by respiration and decomposition.
 Oxygen Cycle	O ₂ released by plants during photosynthesis → Used by animals for respiration → CO ₂ released → Used again by plants.
 Water Cycle	Water evaporates from water bodies and plants → Condenses in clouds → Falls as rain → Collected by rivers, lakes and soil.

7 DECOMPOSITION AND NUTRIENT CYCLE



8 COMPONENTS OF AN ECOSYSTEM – ROLES AND IMPORTANCE

Aspect	Producers (Autotrophs)	Consumers (Heterotrophs)	Decomposers
Role	Make their own food using sunlight, water, CO ₂ and minerals.	Obtain food by eating plants or other animals.	Break down dead organisms and wastes.
Food Source	Inorganic (CO ₂ , water, minerals, sunlight)	Organic (plants, animals or both)	Dead organic matter (plants, animals, wastes)
Examples	Green plants, algae, phytoplankton	Herbivores, Carnivores, Omnivores, Humans	Bacteria, Fungi, Earthworms
Importance	Provide food and oxygen; base of all food chains.	Transfer energy in the food chain; control population.	Recycle nutrients; keep environment clean; maintain soil fertility.

9 EFFECTS WHEN ECOLOGICAL BALANCE IS DISTURBED

- **Extinction of species:** Some plants or animals may disappear.
 - **Increase in pests:** Overgrowth of certain species due to lack of natural control.
 - **Pollution and waste build-up:** Harmful for living organisms.
 - **Disturbed food chains:** Breaks the flow of energy.
 - **Loss of soil fertility and water quality.**
 - **Climate imbalance and extreme weather events.**
- Disturbance in ecological balance affects life on Earth!**

QUICK RECALL

- Decomposers = Nature's recyclers.
- They break down dead matter and return nutrients.
- Nutrient cycles keep elements moving in nature.
- Balance among components keeps ecosystem stable.

KEY IDEA

Decomposers ensure nutrient recycling, while nutrient cycles and the balance among all components ensure the continuity of life on Earth.

EXAM TIPS

- Define decomposers and give examples.
- Draw decomposition cycle and label it.
- Explain carbon, oxygen and water cycles in simple words.
- Write effects of disturbed ecological balance.

CENTUM FOCUS

- Learn definitions word-for-word.
- Draw neat, labelled diagrams.
- Distinguish roles of producers, consumers and decomposers.
- Practice tables and flow charts.
- Use comparison tables in answers.

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SECTION E - BIODEGRADABLE AND NON-BIODEGRADABLE SUBSTANCES

1 MEANING OF BIODEGRADABLE SUBSTANCES

- Substances that can be broken down naturally by microorganisms (bacteria, fungi, actinomycetes) into simpler, harmless substances.
- They return to nature through natural cycles.

2 MEANING OF NON-BIODEGRADABLE SUBSTANCES

- Substances that cannot be broken down or are broken down extremely slowly by microorganisms.
- They persist in the environment for a very long time and cause pollution.

3 EVERYDAY EXAMPLES

Biodegradable Substances

- Food leftovers (rice, roti, fruits)
- Vegetable and fruit peels
- Paper, cardboard
- Cow dung, leaves, grass
- Cotton, wool, jute
- Wood, sawdust

Non-biodegradable Substances

- Plastic, polythene bags
- Glass bottles and jars
- Metal cans, aluminium foils
- Thermocol (expanded polystyrene)
- Synthetic rubber
- Pesticides, DDT, fertiliser residues
- Nylon, polyester fabrics

4 WHY BIODEGRADABLE WASTES DO NOT PERSIST FOR LONG

- They are broken down by decomposers such as bacteria, fungi and actinomycetes.
- Decomposition converts them into simple and harmless substances like carbon dioxide, water, mineral salts and humus.
- These substances are absorbed by soil or plants and recycled in nature.

5 WHY NON-BIODEGRADABLE WASTES ACCUMULATE IN THE ENVIRONMENT

- They are not easily decomposed by natural processes.
- They are resistant to microorganisms, heat, water and chemical reactions.
- Continuous addition of such wastes increases their quantity over time.
- They remain for years or even centuries, causing pollution.

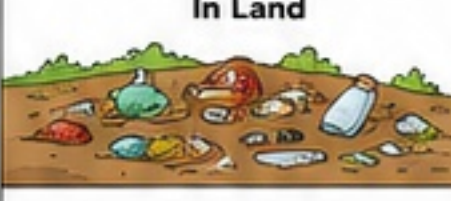
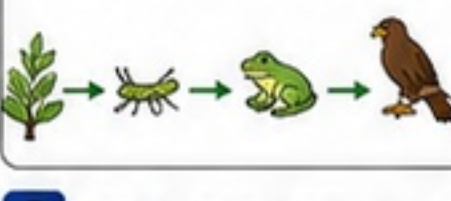
6 HAZARDS OF COMMON NON-BIODEGRADABLE SUBSTANCES

	Plastic & Polythene Clog drains, cause waterlogging, harm animals when eaten, release toxic chemicals when burnt.
	Glass Does not decompose; broken glass causes injuries; may concentrate sunlight and start fires.
	Metal Cans (Aluminium, Tin, Steel) Do not decay; may leak chemicals; mining and processing metals consume resources and energy.
	Pesticides (DDT, BHC, etc.) Toxic to living organisms; cause poisoning, cancer, hormonal imbalance; persist in soil and water.

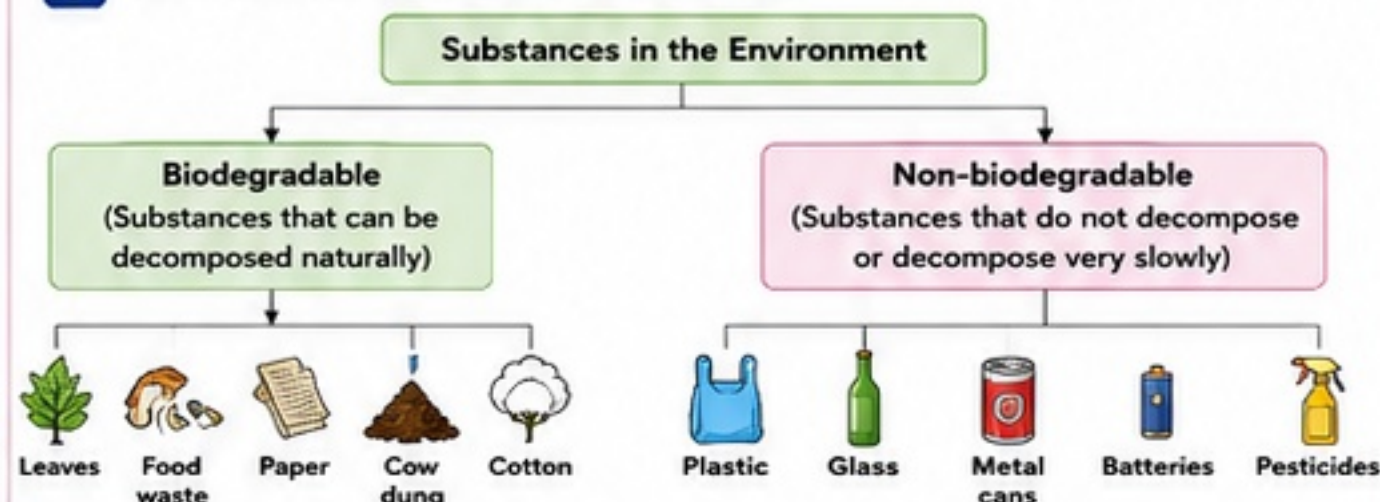
7 DIFFERENCE BETWEEN NATURAL AND SYNTHETIC MATERIALS IN DEGRADATION

	Natural Materials	Synthetic Materials
Origin	From plants and animals	Made by humans in factories
Composition	Complex but natural organic compounds	Man-made chemicals and polymers
Action of Microorganisms	Easily attacked and decomposed	Difficult to attack; very slow or no decomposition
Examples	Leaves, food, cotton, wood, paper, dung	Plastic, nylon, glass, metals, synthetic fibre
Time to Degrade	Short (days to months)	Very long (years to centuries)
Final Products	Simple, harmless substances	Remain as waste or break into harmful fragments
Impact on Environment	Do not accumulate; support natural cycles	Accumulate and cause pollution

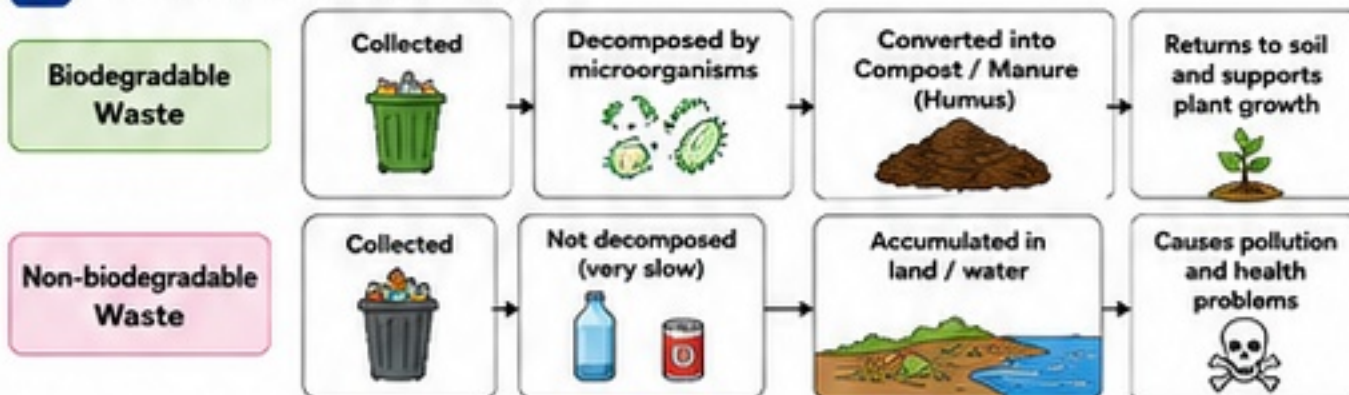
8 HARM CAUSED BY ACCUMULATION

In Land 	- Blocks pores of soil and reduces fertility. - Harmful chemicals leach into soil. - Growth of plants is affected.
In Water 	- Pollutes rivers, lakes and oceans. - Harms aquatic organisms. - Plastic ingestion causes death in fish, turtles and birds.
In Food Chains 	- Harmful substances (pesticides, heavy metals) enter food chains. - They get magnified at higher trophic levels (biomagnification). - Affects health of humans and animals.

9 CLASSIFICATION AND EXAMPLES



10 FROM WASTE TO IMPACT



WHY SOME WASTES ARE PROBLEMATIC

- Not easily decomposed
- Toxic in nature
- Harm living organisms
- Disrupt natural cycles
- Accumulate over time
- Difficult to dispose



QUICK RECALL

- Biodegradable = can be decomposed.
- Non-biodegradable = cannot or very slow.
- Decomposition returns nutrients to nature.
- Accumulation causes pollution and health hazards.



KEY IDEA

Nature can handle wastes it can break down. What it cannot break down, we must reduce, reuse and recycle to protect our environment and ourselves.



EXAM TIPS

- Define biodegradable and non-biodegradable substances.
- Give examples of each.
- State reasons for accumulation and their hazards.
- Draw diagrams to show impacts and solutions.



CENTUM FOCUS

- Learn definitions word-for-word.
- Compare natural vs synthetic materials.
- Practise tables & flow charts.
- Write causes, effects and solutions in points.
- Revise key terms daily.



Understand nature, reduce waste, make wise choices — for a clean, safe and healthy future!



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SECTION F - WASTE MANAGEMENT, 3Rs/5Rs AND ENVIRONMENTAL RESPONSIBILITY

1 NEED FOR WASTE MANAGEMENT

- Our daily activities produce waste.
- Improper disposal pollutes air, water and soil.
- It causes health problems, foul smell and blocks drains.
- Proper waste management protects environment and our health.

2 SEGREGATION OF WASTE AT SOURCE

- Waste should be separated at the place where it is produced (home, school, office).
- Wet (biodegradable) waste and Dry (non-biodegradable) waste must be kept in different bins.

3 BIODEGRADABLE WASTE HANDLING

- **Composting** – Decomposition of organic waste by microorganisms in presence of air to form compost (manure).
- **Vermicomposting** – Earthworms break down organic waste and form nutrient-rich vermicompost.
- **Biogas** – Organic waste is decomposed in absence of air to produce biogas used as fuel.

4 NON-BIODEGRADABLE WASTE HANDLING

- **Recycling** – Converting used materials (paper, glass, plastic, metal) into new products.
- **Reuse** – Using items again for the same or different purpose.
- **Safe Disposal** – Non-recyclable waste should be disposed in sanitary landfill or incinerated under controlled conditions.

5 3Rs AND 5Rs

- 3Rs – Reduce, Reuse, Recycle.
- Additional 2Rs – Refuse, Recover.
- Together they form 5Rs for a sustainable life.

6 COMPOST PIT

A simple pit dug in the ground or a container where biodegradable waste is buried and decomposed by microorganisms to form compost (manure).

7 LANDFILL AND SANITARY LANDFILL

- **Landfill** – Waste is buried in open land.
- **Sanitary Landfill** – Waste is buried in a well-designed site with a clay/plastic lining, covered daily with soil to prevent pollution and pests.

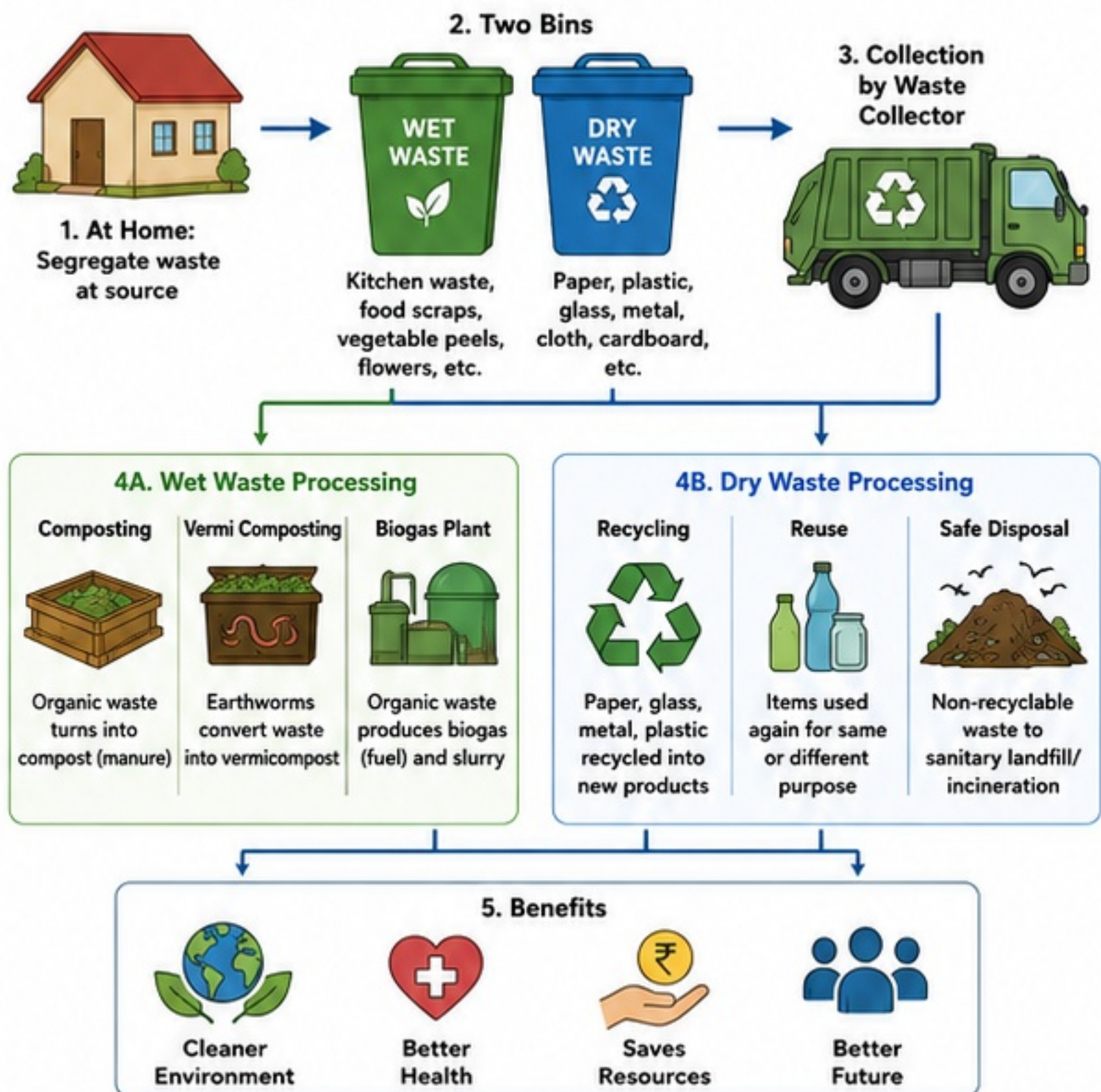
8 INCINERATION

- Burning of waste at high temperature in an incinerator to reduce its volume.
- **Limitations** – Releases gases/ash if not controlled; costly; not suitable for plastics containing harmful chemicals.

9 OUR RESPONSIBILITY

- **Individual** – Segregate waste, follow 5Rs.
- **Community** – Clean surroundings, awareness drives, compost pits, proper collection.
- **School** – Eco clubs, waste audits, recycling projects, tree planting, no-litter policy.

10 WASTE SEGREGATION AND MANAGEMENT – FROM HOME TO DISPOSAL



11 THE 5Rs – MEANING AND EXAMPLES

5R	Meaning	What it Means	Examples in Daily Life
Refuse	Say NO to what you don't need	Avoid items that create waste.	Refuse plastic straws, free gift wrappers.
Reduce	Use less	Minimise consumption and waste.	Carry cloth bag, buy only what is needed.
Reuse	Use again	Use items again for same or different purpose.	Reuse bottles, boxes, old notebooks.
Recycle	Convert and use again	Convert used materials into new products.	Recycle paper, glass, metals, plastics.
Recover	Recover materials or energy	Recover useful materials or energy from waste.	Biogas from waste, metal recovery.

12 COMPOSTING vs RECYCLING

Aspect	Composting	Recycling
Type of Waste	Biodegradable (organic)	Non-biodegradable (inorganic)
Process	Microorganisms break down waste	Materials are processed and remade
Product	Compost (manure)	New products
Time Taken	Weeks to months	Varies (days to weeks)
Use	Improves soil fertility	Saves raw materials and energy

13 ECO-FRIENDLY HABITS CHECKLIST

- ☐ I segregate wet and dry waste at home/school.
- ☐ I follow the 5Rs in my daily life.
- ☐ I avoid single-use plastics.
- ☐ I use both sides of paper.
- ☐ I carry a cloth bag and water bottle.
- ☐ I compost kitchen waste (if possible).
- ☐ I keep my surroundings clean.
- ☐ I save water and electricity.
- ☐ I plant trees and care for plants.
- ☐ I spread awareness among others.

QUICK RECALL

- Waste management keeps our environment clean.
- Segregate – Wet and Dry.
- Handle waste: Compost, Vermicompost, Biogas, Recycle, Reuse, Safe Disposal.
- Follow 5Rs: Refuse, Reduce, Reuse, Recycle, Recover.

KEY IDEA

Waste is not useless – it is a resource. Manage it well, and it becomes wealth for us and the planet.

EXAM TIPS

- Define terms clearly (e.g., compost pit, landfill).
- Draw neat flow diagrams for waste management.
- Write differences in points.
- Give examples of 5Rs.
- Use comparison tables in answers.

CENTUM FOCUS

- Learn definitions word-for-word.
- Practice diagrams and flow charts.
- Write points in proper order.
- Revise tables daily.
- Solve previous year questions.

★ Think before you throw! Manage waste today for a cleaner, healthier tomorrow. ★

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SECTION G – OZONE LAYER, OZONE DEPLETION AND UV RADIATION

1 WHAT IS OZONE?

Ozone (O_3) is a gas made up of three oxygen atoms. It has a sharp smell and is bluish in colour. It is different from oxygen (O_2).

2 WHERE IS OZONE LAYER FOUND?

The ozone layer is found in the stratosphere, about 15–35 km above the Earth's surface. It forms a thin protective shield around our planet.

3 ROLE OF THE OZONE LAYER

- Absorbs most of the Sun's harmful ultraviolet (UV) rays, especially UV-B and UV-C.
- Protects living organisms from damage.
- Helps maintain life on Earth.

4 OZONE FORMATION IN THE ATMOSPHERE

Ozone is continuously formed and broken down naturally in the stratosphere.

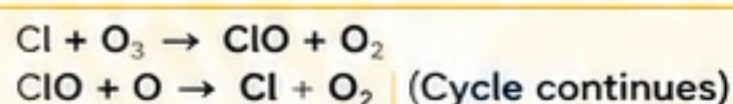
- UV-C rays break oxygen molecule (O_2) into oxygen atoms (O).
- $O + O_2 \rightarrow O_3$ (Ozone)
- UV-B rays break O_3 back into $O_2 + O$.
- A natural balance keeps ozone layer stable.

5 OZONE DEPLETION – MEANING

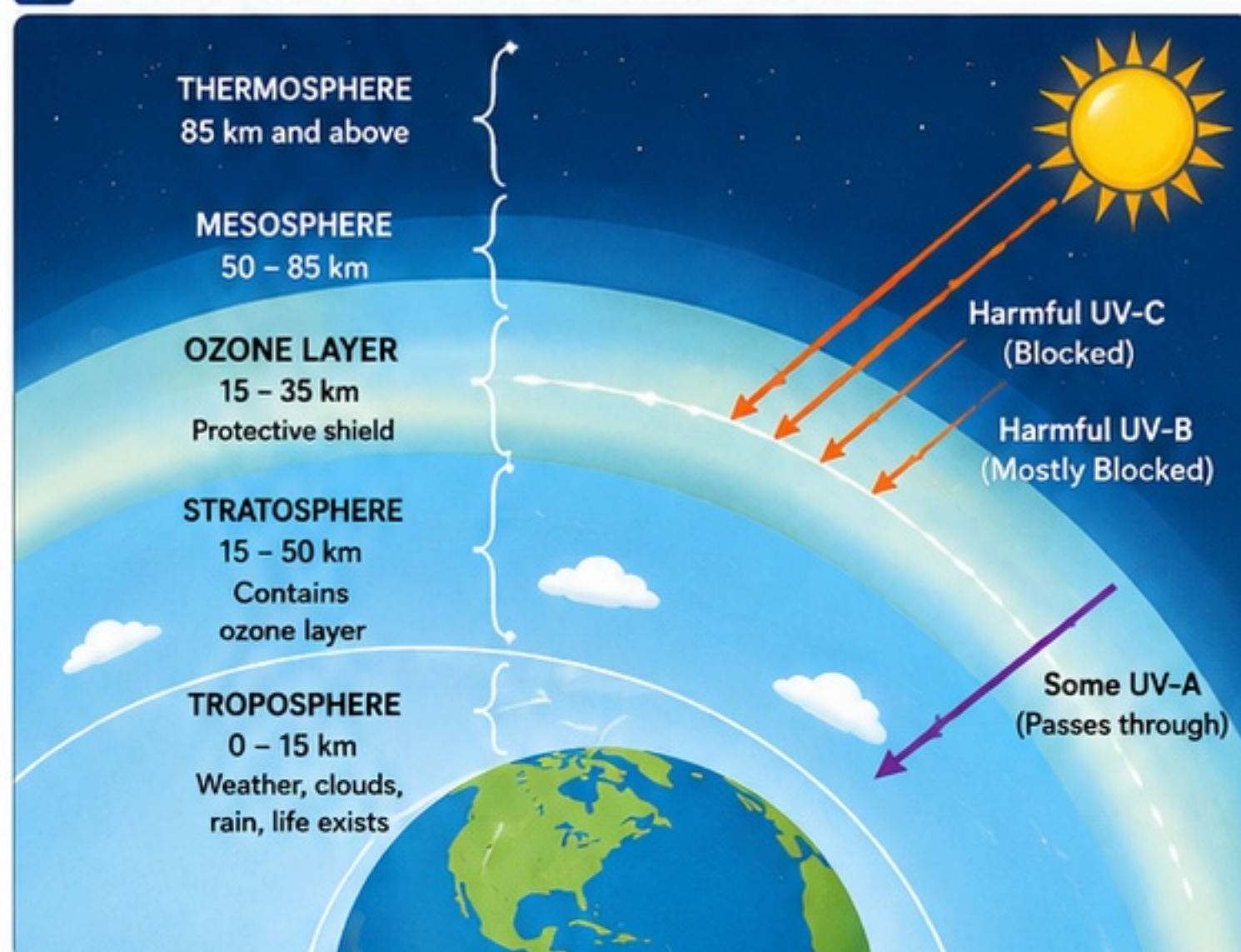
Ozone depletion is the gradual thinning or reduction of ozone layer in the stratosphere. It allows more harmful UV rays to reach Earth.

6 CAUSE – CFCs AND SIMILAR CHEMICALS

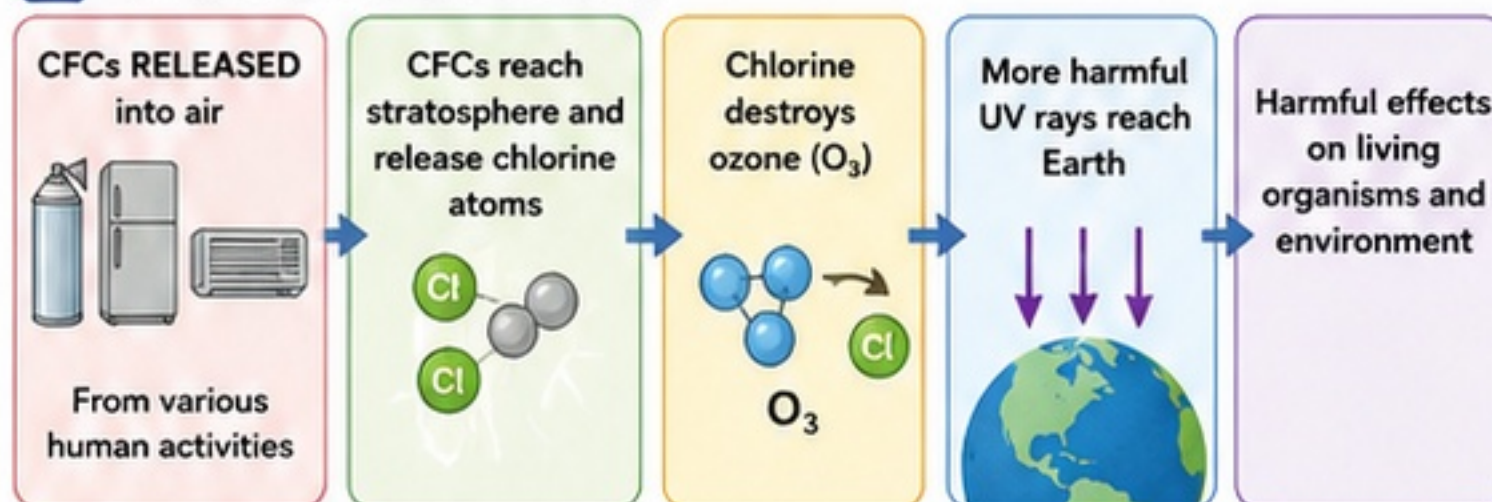
- Chlorofluorocarbons (CFCs) and related chemicals are main causes of ozone depletion.
- They are stable, non-flammable and non-toxic, but damage ozone.
- In the stratosphere, UV rays break CFCs and release chlorine (Cl) atoms.
- Cl atoms react with ozone and destroy it.



7 STRUCTURE OF ATMOSPHERE AND OZONE LAYER



8 CAUSE-EFFECT FLOW CHART



9 SOURCES OF CFCs



12 IMPORTANT TABLE

Aspect	Ozone Layer Importance	Causes of Depletion	Harmful Effects	Preventive Measures
What	Protects Earth from harmful UV rays	CFCs, Halons, Carbon tetrachloride, Methyl chloroform, etc.	Skin cancer, cataract, weak immune system, plant & aquatic damage	Reduce CFCs, use eco-friendly products, recycle, spread awareness
Where	Stratosphere (15–35 km above Earth)	Human activities releasing ODS*	Health problems, crop loss, ecosystem imbalance	Follow laws, Montreal Protocol, save ozone, save life
Why	Maintains life, balance and health of ecosystems	ODS remain stable and reach stratosphere	More UV rays cause long-term damage	Every small step helps to protect our future

*ODS – Ozone Depleting Substances

- 10 Health problems:** More UV rays can cause skin cancer, premature ageing, sunburn and cataracts (eye damage).
- ② Weak immune system:** UV radiation can weaken the body's ability to fight diseases.
- ③ Damage to plants:** Affects growth, reduces crop yield and damages forest plants.
- ④ Harm to aquatic life:** Affects plankton and young stages of fish, disturbing the aquatic food chain.

11 WAYS TO PROTECT THE OZONE LAYER

- Reduce and stop use of CFCs, halons and other – ozone-depleting substances. Use eco-friendly alternatives in refrigerators, air conditioners and sprays.
- Proper maintenance and recycling of old cooling appliances.
- Avoid unnecessary aerosol sprays.
- Use recyclable and biodegradable packaging materials.
- Follow international agreements like the Montreal Protocol.
- Spread awareness – "Save Ozone, Save Life!"

QUICK RECALL

- Ozone (O_3) in stratosphere absorbs harmful UV rays.
- CFCs destroy ozone.
- Ozone depletion harms health, plants and aquatic life.
- Save ozone, save life!

KEY IDEA

Ozone layer is nature's protective umbrella. Let's protect it by reducing CFCs and choosing eco-friendly alternatives.

EXAM TIPS

- Learn location and role of ozone layer.
- Remember CFCs as main cause.
- Write effects in health, plants and aquatic life.
- Use flow chart or table in answers.

CENTUM FOCUS

- Learn definitions word-for-word.
- Draw labelled diagram of atmosphere.
- Practice tables and flow charts.
- Write points in systematic order.
- Revise daily for better recall.



Save Ozone today – for a safer, healthier and brighter tomorrow!



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SECTION H – BIOLOGICAL MAGNIFICATION (BIOMAGNIFICATION) AND ENVIRONMENTAL HAZARDS

1 MEANING OF BIOLOGICAL MAGNIFICATION

Biological magnification (biomagnification) is the gradual increase in the concentration of non-biodegradable toxic substances in living organisms at successive trophic levels of a food chain.

2 WHY IT HAPPENS (CAUSE)

- Non-biodegradable toxic substances such as pesticides (e.g., DDT), PCBs and heavy metals (e.g., mercury, lead) are not broken down by natural processes.
- They persist in the environment for a long time.
- They are absorbed by organisms and stored in body fats and tissues.
- With each step up the food chain, these toxins accumulate and their concentration increases.

3 ENTRY OF PESTICIDES INTO FOOD CHAINS

- Pesticides are sprayed on crops to kill pests.
- Rain washes them into soil and water.
- They are absorbed by plants and taken up by small organisms in water or soil.
- These organisms are eaten by small fish/animals.
- Larger fish, birds and animals eat the smaller ones, thus the pesticides enter the food chain.

4 INCREASE IN CONCENTRATION AT SUCCESSIVE TROPHIC LEVELS

- Each organism absorbs the toxin from the organisms it eats.
- The toxin is not easily excreted.
- So, its concentration becomes higher and higher at each trophic level.
- Top level consumers have the maximum concentration.

5 DDT – A CLASSIC EXAMPLE

- DDT (Dichloro-Diphenyl-Trichloroethane) is a persistent organochlorine pesticide.
- It does not degrade easily and enters food chains.
- Its concentration increases a million times from water to top predators.
- DDT caused eggshell thinning in birds of prey like eagles, falcons and ospreys, leading to decline in their populations.

BIODEGRADABLE WASTES vs NON-BIODEGRADABLE TOXIC POLLUTANTS

Biodegradable Wastes	Non-biodegradable Toxic Pollutants
Broken down by natural processes (bacteria, fungi).	Cannot be broken down easily; remain for long time.
Do not accumulate in organisms.	Accumulate in organisms and build up in food chains.
Do not show biomagnification.	Show biomagnification; concentration increases at higher trophic levels.
Examples: Vegetable wastes, paper, sewage.	Examples: DDT, PCBs, mercury, lead, cadmium.

QUICK RECALL

- Biomagnification = Increase in toxin concentration up the food chain.
- Cause = Non-biodegradable toxics.
- Example = DDT.
- Affects top consumers most.

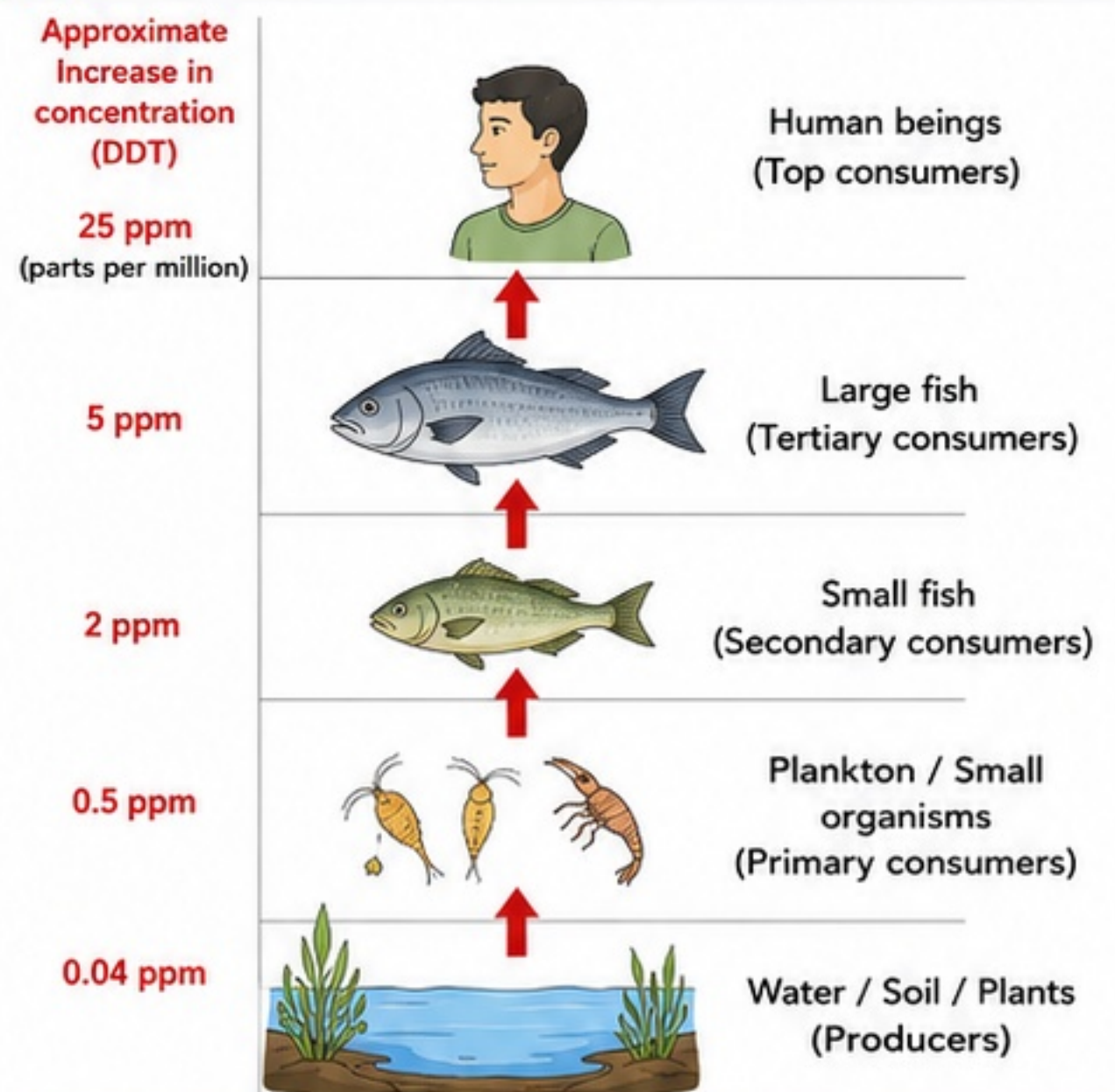
KEY IDEA

Toxins move from environment → plants / small organisms → larger animals → top predators, and get magnified.

6 WHY HUMAN BEINGS AND TOP CARNIVORES ARE BADLY AFFECTED

- Humans and top carnivores are at the highest trophic level.
- They eat many organisms, so they accumulate the maximum amount of toxins.
- Toxins disturb their growth, reproduction, nervous system, hormones and can cause cancers and death.

7 BIOMAGNIFICATION IN A TYPICAL FOOD CHAIN



Note: ppm = parts per million. Concentration increases many times from the environment to the top consumers.

8 ENVIRONMENTAL HAZARDS CAUSED BY TOXIC CHEMICALS

Effects of Toxic Chemicals

	Reproductive problems in birds Eggshell thinning, poor hatching, deformities, decline in populations.
	Human health risks Cancers, hormone disorders, nervous system damage, birth defects, kidney and liver problems.
	Contamination of water and soil Pollutes lakes, rivers, groundwater and soil for long time.
	Danger to top predators Top carnivores (eagles, hawks, sharks, etc.) accumulate maximum toxins and may die or become sterile.

EXAM TIPS

- Define biomagnification clearly.
- Draw a food chain and show increase in concentration.
- Give example (DDT or mercury).
- Mention effects on birds, humans and environment.

CENTUM FOCUS

- Understand the reason for increase in concentration.
- Practice food chain diagrams.
- Learn real-life examples.
- Write points in systematic order in answers.



Be aware, be responsible — reduce use of chemicals and protect all forms of life!



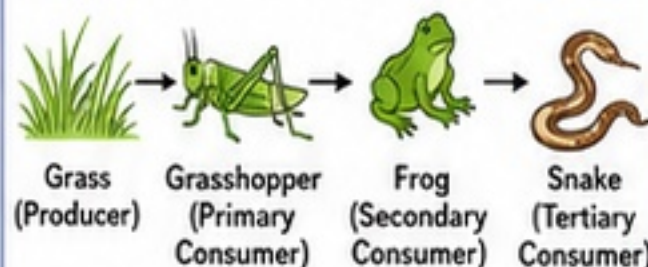
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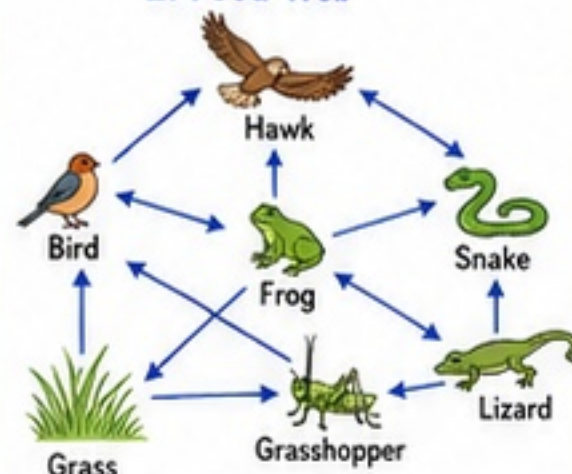
SECTION I - IMPORTANT DIAGRAMS, COMPARISON TABLES, ANSWER FRAMES AND PRACTICE SET

1 MUST-KNOW DIAGRAMS

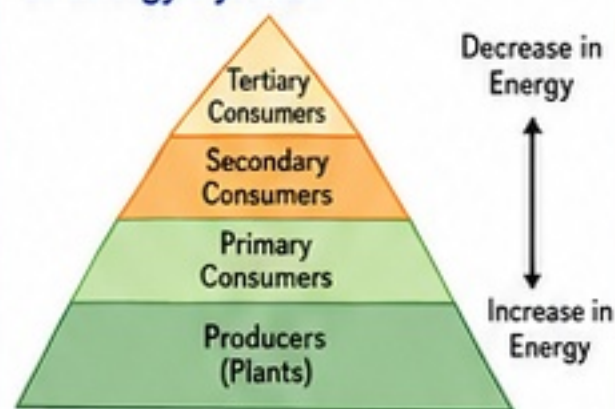
1. Simple Food Chain



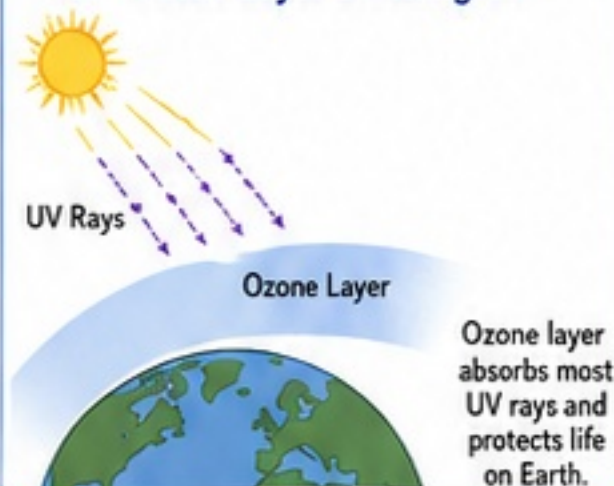
2. Food Web



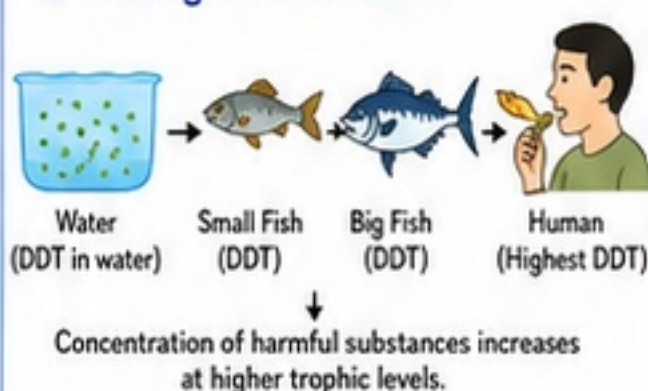
3. Energy Pyramid



4. Ozone Layer Blocking UV



5. Biomagnification Chain



6. Waste Segregation



2 SHORT-ANSWER (BOARD STYLE) – ANSWER FRAMES

1 Define ecosystem.

- An ecosystem is _____.

2 What is a food chain?

- A food chain is _____.

3 State the 10% law.

- According to the 10% law, _____.

4 Differentiate between biodegradable and non-biodegradable substances.

- Biodegradable substances _____ whereas non-biodegradable substances _____.

5 What is biological magnification?

- Biological magnification is _____.

6 Why is the ozone layer important?

- The ozone layer is important because _____.

3 COMPARISON TABLES

1. Food Chain vs Food Web

Basis	Food Chain	Food Web
Definition	A single linear pathway of energy transfer.	A network of interlinked food chains.
Structure	Simple and linear.	Complex and interdependent.
Stability	Less stable; easily disturbed.	More stable; organisms have alternative food sources.
Example	Grass → Grasshopper → Frog → Snake	Interconnected system of multiple food chains.

2. Biodegradable vs Non-biodegradable

Basis	Biodegradable	Non-biodegradable
Definition	Can be broken down naturally by microorganisms.	Cannot be broken down or take a very long time.
Examples	Food waste, Leaves, Paper, Cotton	Plastic, Glass, Metals, Synthetic fibres
Effect on Environment	Environment-friendly; recycles nutrients.	Pollutes environment; accumulation causes harm.
Disposal	Can be composted.	Should be recycled or safely disposed.

3. Bioaccumulation vs Biomagnification

Basis	Bioaccumulation	Biomagnification
Definition	Build-up of harmful substances in an organism over time.	Increase in concentration of harmful substances at successive trophic levels.
Where It Occurs	Within a single organism.	Across food chain from lower to higher organisms.
Example	DDT accumulating in the fat tissues of a fish.	DDT concentration increasing from water → fish → bird → human.
Cause	Continuous exposure from environment.	Transfer of toxins through eating contaminated food.
Result	Harm to individual organism.	Greater harm to top consumers.

4 PRACTICE SET (10 MARKS)

A. One-line Questions (Answer in one line)

(5 × 1 = 5)

- What is an ecosystem?
- Name the non-living components of an ecosystem.
- What is decomposer?
- Give one example of a biodegradable substance.
- Which gas in the atmosphere causes global warming?

B. Short Answer Questions (Answer in 2–3 lines)

(5 × 1 = 5)

- State the 10% law.
- What is the importance of decomposers in an ecosystem?
- Differentiate between food chain and food web.
- What is biomagnification? Give an example.
- How can we protect the ozone layer?

SELF-CHECK ANSWERS

- Functional unit of nature.
- Air, Water, Soil, Sunlight, Temperature.
- Organism that breaks down dead matter.
- Paper.
- Carbon dioxide.
- Only 10% energy is passed to the next trophic level.
- They decompose dead matter and recycle nutrients.
- Food chain – linear; Food web – interlinked.
- Increase of toxins in food chain; e.g., DDT in water – fish – bird – human.
- Avoid CFCs, use eco-friendly products, follow laws.

QUICK RECALL

- Ecosystem = Functional unit of nature.
- Energy flow is one-way.
- Decomposers = Cleaners of nature.
- Protect ozone, protect life.

KEY IDEA

- Everything in nature is connected.
- Balance in nature = Sustainable future.
- Reduce, Reuse, Recycle – Your 3R rule.

EXAM TIPS

- Draw neat labelled diagrams for full marks.
- Write definitions in simple and clear words.
- Use comparison tables for differences.
- Give examples wherever possible.

CENTUM FOCUS

- Learn definitions word-for-word.
- Practice diagrams daily.
- Write points in systematic order.
- Revise tables and answer frames regularly.



Understand nature, respect it, protect it – because a balanced environment ensures a better future!



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SECTION J - MASTER REVISION SHEET, NCERT TRAPS AND CENTUM FOCUS

1 KEY DEFINITIONS

Term	Meaning
Environment	All the surrounding living and non-living things that influence the life and activities of organisms.
Ecosystem	A functional unit of nature where living organisms (biotic) interact with non-living components (abiotic) and exchange matter and energy.
Food chain	A linear sequence of organisms in which each is eaten by the next and energy flows in one direction.
Food web	A network of interconnected food chains.
Trophic level	Each step or position in a food chain or food web.
Producer	Autotrophs that make their own food using sunlight (e.g., green plants, algae).
Consumer	Heterotrophs that cannot make their own food and depend on other organisms for food.
Decomposer	Organisms that break down dead plants, animals and wastes into simpler substances and return nutrients to the environment.
Biodegradable	Substances that can be broken down naturally by decomposers.
Non-biodegradable	Substances that cannot be decomposed easily and persist in the environment.
Biomagnification	Increase in concentration of non-biodegradable toxic substances at higher trophic levels.
Ozone layer	A layer of ozone in the upper atmosphere (stratosphere) that absorbs harmful UV radiation from the Sun and protects life on Earth.

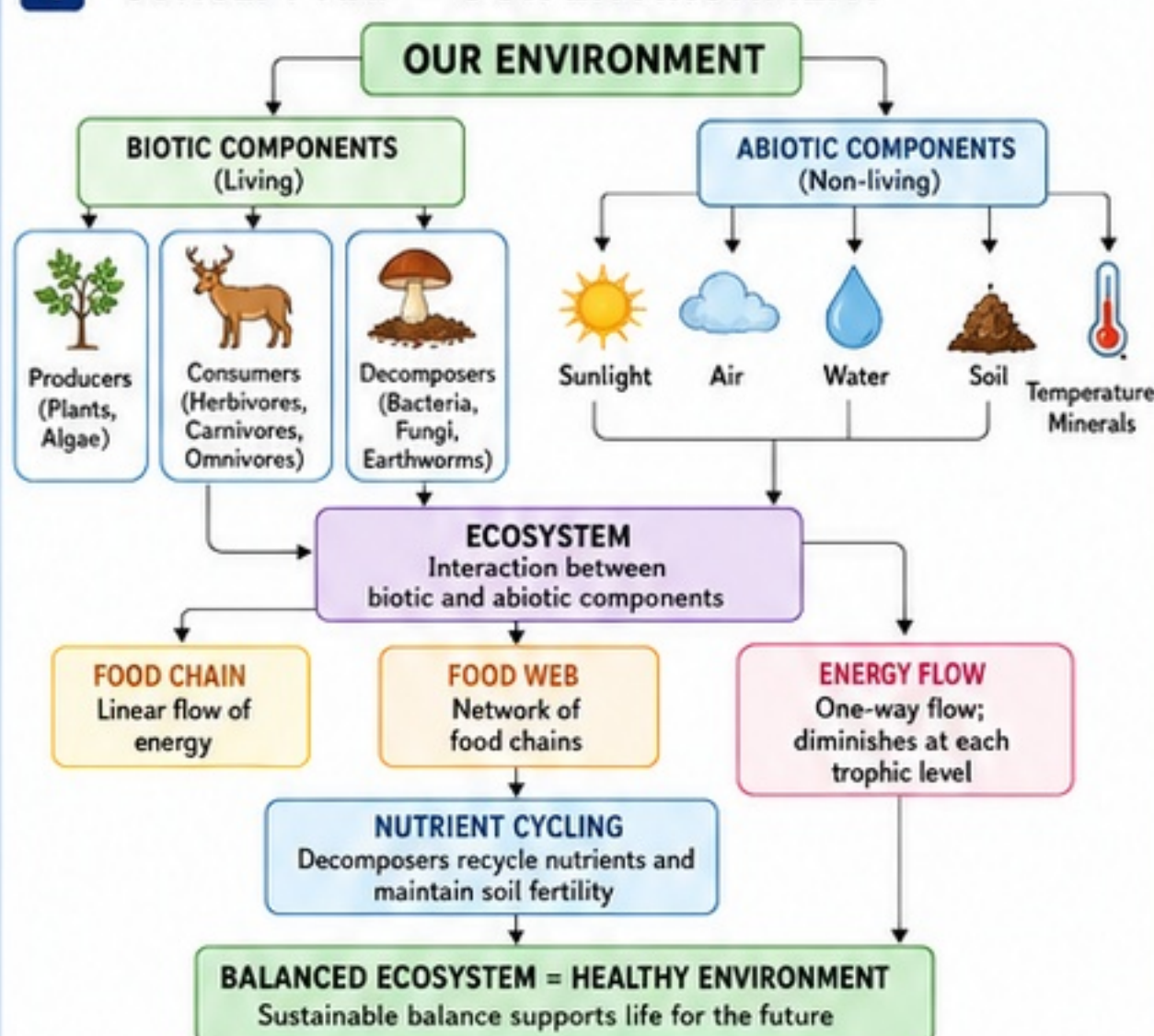
2 MASTER CHAPTER SUMMARY

- Our environment includes living (biotic) and non-living (abiotic) components.
- An ecosystem is a functional unit where biotic and abiotic components interact and maintain a balance.
- Biotic components: Producers, Consumers, Decomposers.
- Abiotic components: Air, Water, Soil, Sunlight, Temperature, Minerals.
- Food chains show one-way flow of energy; food webs show interconnections.
- Energy flow in an ecosystem is *unidirectional* and diminishes at each step.
- Decomposers recycle nutrients and maintain soil fertility.
- Non-living factors affect distribution, growth and survival of organisms.
- Pollution (air, water, soil) harms living organisms and ecosystems.*
- Ozone layer in the stratosphere protects life by absorbing UV rays.*
- Biomagnification increases concentration of harmful chemicals in higher trophic levels and threatens top consumers including humans.*
- Sustainable environment ensures balance and supports life for the future.*

3 NCERT TRAPS / COMMON MISCONCEPTIONS

Trap / Misconception	Why It's Wrong	Correct Explanation
Arrows in a food chain show the direction an organism eats.	Many think arrow points from prey to predator because of eating.	Arrows show the flow of energy from food to consumer.
All wastes are biodegradable.	Many assume everything can be decomposed naturally.	Only some wastes (e.g., vegetable peels) are biodegradable. Plastics, glass, metals are not.
Ozone is harmful in the atmosphere.	Ozone is associated with air pollution.	Ozone at ground level is harmful, but in the upper atmosphere (ozone layer) it protects life.
Biomagnification happens for biodegradable substances.	Some think all toxins increase up the chain.	Biomagnification occurs only with non-biodegradable, persistent, fat-soluble toxic substances.
Decomposers are consumers.	Decomposers break down dead matter.	They are a separate group that recycle nutrients and are not eaten by others.

4 CONCEPT MAP – OUR ENVIRONMENT



5 10 MUST REMEMBER POINTS

- Environment = Biotic (living) + Abiotic (non-living).
- Ecosystem = Functional unit with interaction and balance.
- Producers make food; consumers depend on others.
- Decomposers recycle nutrients and maintain soil fertility.
- Food chain = Linear; Food web = Interconnected.
- Energy flows in one direction and decreases at each step.
- Non-living factors affect life and distribution of organisms.
- Pollution disturbs balance and harms ecosystems.*
- Ozone layer (stratosphere) protects from harmful UV rays.*
- Biomagnification increases toxins in higher trophic levels.*

6 CENTUM CHECKLIST – ARE YOU READY?

Task	Done (✓)
I can define all key terms and use them correctly.	☐
I can draw and label a food chain and a food web.	☐
I can explain the flow of energy in an ecosystem.	☐
I can list and classify biotic and abiotic components.	☐
I can explain the role of each component in an ecosystem.	☐
I can explain pollution, ozone layer and biomagnification.	☐
I can solve all NCERT questions from this chapter.	☐
I have revised all tables, diagrams and summary points.	☐
I have practised HOTS / Case-based / Assertion-Reason questions.	☐

TARGET: FULL MARKS!

7 BEFORE EXAM – QUICK GLANCE

CONCEPTS	DIAGRAMS	TABLES	KEY TERMS	FORMULAS/FACTS	TIPS
- Environment - Ecosystem - Components - Interaction - Balance	- Food chain - Food web - Ecosystem components - Decomposer cycle	- Biotic vs Abiotic - Producers, Consumers, Decomposers - Pollutants - Biodegradable vs Non-biodegradable	- Producer - Consumer - Decomposer - Trophic level - Biomagnification - Ozone layer	- Energy flow = One-way - Energy loss at each step - Ozone protects from UV rays - Pollution = Damage	- Read questions carefully - Use keywords in answers - Draw diagrams neatly - Write points clearly

★ Understand nature, respect it, protect it — because a balanced environment ensures a better future! ★